



CHANDIGARH UNIVERSITY

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Role of Interfaces and Enums in Software Design.

Roles of Interfaces

- Define contracts without Implementation
- Promote abstraction and polymorphism
- Enables loose coupling b/w components
- Supports dependency inversion principle (DIP)
- Allow multiple implementations without changing client code.

Example :-

- PaymentService interface with method `pay()`
- Implementations :
 - Credit Card Payment
 - UPI Payment
- Client depends on interface, not concrete class.

Roles of Enums

- Represent a fixed set of constants
- Improve type safety.
- Replace magic strings or numbers.
- Make code more readable and maintainable.
- Useful for states, roles, statuses.

Example :-

- `PaymentStatus { SUCCESS, FAILED, PENDING }`
- Prevents invalid status values.

How Interface Enable Loose Coupling

Loose Coupling - Components depend on abstractions, not implementations

- Change in one module don't affect others
- Easier testing, maintenance and extension

How Interfaces Helps ?

- Client code references interface type
- Implementation can be swapped at runtime
- New features added without modifying existing code.
- Supports plug-and-play architecture.

e.g.

- Interface : NotificationService
- Implementations : EmailNotification
SMSNotification
- OrderService depends only on NotificationService

Benefits

- High Flexibility
- Better testability (mock implementations)
- Reduced dependency ripple effects
- Follows Open - Closed Principle

Conclusion

- Interfaces are the key tool for achieving loose coupling.
- Widely used in enterprise and microservices architecture